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Nonlinear Control of a Quadrotor using Backstepping and Lyapunov Stability

1 Introduction

This report presents the nonlinear control design of a quadrotor based on rigid body dynamics and Lyapunov stability theory. The controller is divided into three cascaded subsystems:

- Attitude controller
- Altitude controller
- Position controller

Two simulations were conducted:

1. Set-point stabilization
2. Circular trajectory tracking

2 System Parameters

$$m = 1.336 \text{ kg}$$

$$g = 9.80665 \text{ m/s}^2$$

$$I_x = 0.0259 \text{ kg}\cdot\text{m}^2$$

$$I_y = 0.0260 \text{ kg}\cdot\text{m}^2$$

$$I_z = 0.0397 \text{ kg}\cdot\text{m}^2$$

Controller gains:

$$k_1 = k_2 = k_3 = k_4 = 2.5$$

$$k_5 = k_6 = 2$$

$$k_7 = 2.2$$

$$k_8 = 0.5$$

$$k_9 = k_{11} = 0.4$$

3 Quadrotor Dynamics

Let the generalized coordinates be

$$\boldsymbol{\xi} = (x, y, z, \phi, \theta, \psi)$$

Body angular velocities:

$$\boldsymbol{\omega} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

3.1 Translational Dynamics

The translational acceleration is obtained from

$$m\ddot{\mathbf{p}} = R(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ U_1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix}$$

where $R(\phi, \theta, \psi)$ is the ZYX rotation matrix.

The third column of the rotation matrix is

$$R_3 = \begin{bmatrix} c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\phi s_\theta s_\psi - s_\phi c_\psi \\ c_\phi c_\theta \end{bmatrix}$$

Thus,

$$\ddot{x} = \frac{U_1}{m}(c_\phi s_\theta c_\psi + s_\phi s_\psi) \quad (1)$$

$$\ddot{y} = \frac{U_1}{m}(c_\phi s_\theta s_\psi - s_\phi c_\psi) \quad (2)$$

$$\ddot{z} = \frac{U_1}{m}(c_\phi c_\theta) - g \quad (3)$$

Define:

$$\gamma_1 = c_\phi s_\theta c_\psi + s_\phi s_\psi \quad (4)$$

$$\gamma_2 = c_\phi s_\theta s_\psi - s_\phi c_\psi \quad (5)$$

Thus,

$$\ddot{x} = \frac{U_1}{m}\gamma_1, \quad \ddot{y} = \frac{U_1}{m}\gamma_2$$

3.2 Rotational Dynamics

$$I_x \dot{p} = (I_y - I_z)qr + U_2 \quad (6)$$

$$I_y \dot{q} = (I_z - I_x)pr + U_3 \quad (7)$$

$$I_z \dot{r} = (I_x - I_y)pq + U_4 \quad (8)$$

4 Control Laws

4.1 Attitude Controller

Errors:

$$e_1 = \phi - \phi_d \quad (9)$$

$$e_2 = \dot{\phi} - \dot{\phi}_d + k_1 e_1 \quad (10)$$

$$e_3 = \theta - \theta_d \quad (11)$$

$$e_4 = \dot{\theta} - \dot{\theta}_d + k_3 e_3 \quad (12)$$

$$e_5 = \psi - \psi_d \quad (13)$$

$$e_6 = \dot{\psi} - \dot{\psi}_d + k_5 e_5 \quad (14)$$

Control torques:

$$U_2 = I_x \left[-\frac{(I_y - I_z)}{I_x} qr + (k_1^2 - 1)e_1 - (k_1 + k_2)e_2 \right] \quad (15)$$

$$U_3 = I_y \left[-\frac{(I_z - I_x)}{I_y} pr + (k_3^2 - 1)e_3 - (k_3 + k_4)e_4 \right] \quad (16)$$

$$U_4 = I_z \left[-\frac{(I_x - I_y)}{I_z} pq + (k_5^2 - 1)e_5 - (k_5 + k_6)e_6 \right] \quad (17)$$

4.2 Altitude Controller

$$e_7 = z - z_d \quad (18)$$

$$e_8 = \dot{z} - \dot{z}_d + k_7 e_7 \quad (19)$$

$$U_1 = \frac{m}{\cos \phi \cos \theta} [g + (k_7^2 - 1)e_7 - (k_7 + k_8)e_8]$$

4.3 Position Controller

$$e_9 = x - x_d \quad (20)$$

$$e_{10} = \dot{x} - \dot{x}_d + k_9 e_9 \quad (21)$$

$$e_{11} = y - y_d \quad (22)$$

$$e_{12} = \dot{y} - \dot{y}_d + k_{11} e_{11} \quad (23)$$

Virtual inputs:

$$\gamma_1 = \begin{cases} \gamma_{max}, & e_{10} \geq 0 \\ -\gamma_{max}, & e_{10} < 0 \end{cases} \quad \gamma_2 = \begin{cases} \gamma_{max}, & e_{12} \geq 0 \\ -\gamma_{max}, & e_{12} < 0 \end{cases}$$

Derivation of Desired Angles

From the translational dynamics:

$$\ddot{x} = \frac{U_1}{m} \gamma_1, \quad \ddot{y} = \frac{U_1}{m} \gamma_2$$

and from the rotation matrix expansion:

$$\gamma_1 = c_{\phi_d} s_{\theta_d} c_{\psi} + s_{\phi_d} s_{\psi} \quad (24)$$

$$\gamma_2 = c_{\phi_d} s_{\theta_d} s_{\psi} - s_{\phi_d} c_{\psi} \quad (25)$$

(from the rotation matrix)

Solving for ϕ_d :

Multiply the first equation by $\sin \psi$ and subtract the second multiplied by $\cos \psi$:

$$\gamma_1 \sin \psi - \gamma_2 \cos \psi = s_{\phi_d}$$

Thus,

$$\sin \phi_d = \gamma_1 \sin \psi - \gamma_2 \cos \psi$$

Once ϕ_d is known, substitute into the first equation:

$$c_{\phi_d} s_{\theta_d} c_{\psi} = \gamma_1 - s_{\phi_d} s_{\psi}$$

Thus,

$$s_{\theta_d} = \frac{\gamma_1 - \sin \phi_d \sin \psi}{\cos \phi_d \cos \psi}$$

and therefore

$$\theta_d = \arcsin \left(\frac{\gamma_1 - \sin \phi_d \sin \psi}{\cos \phi_d \cos \psi} \right)$$

5 Lyapunov Stability Analysis

5.1 Attitude Subsystem

$$V_1 = \frac{1}{2} \sum_{i=1}^6 e_i^2$$

$$\dot{V}_1 = -k_1 e_1^2 - k_2 e_2^2 - k_3 e_3^2 - k_4 e_4^2 - k_5 e_5^2 - k_6 e_6^2$$

5.2 Altitude Subsystem

$$V_2 = \frac{1}{2}(e_7^2 + e_8^2)$$

$$\dot{V}_2 = -k_7 e_7^2 - k_8 e_8^2$$

5.3 Position Subsystem

$$V_3 = \frac{1}{2}(e_9^2 + e_{10}^2 + e_{11}^2 + e_{12}^2)$$

$$\dot{V}_3 = e_{10}\gamma_1 \frac{U_1}{m} + e_{12}\gamma_2 \frac{U_1}{m} - k_9 e_9^2 - k_{11} e_{11}^2$$

Choosing γ_1 and γ_2 opposite in sign to e_{10} and e_{12} ensures

$$\dot{V}_3 \leq 0$$

6 Simulation Results

6.1 Set-Point Stabilization

The quadrotor was commanded to reach

$$(x_d, y_d, z_d) = (1, 1, 2)$$

All states converged asymptotically and control inputs remained bounded.

6.2 Trajectory Tracking

$$x_d(t) = \sin(0.2t) \tag{26}$$

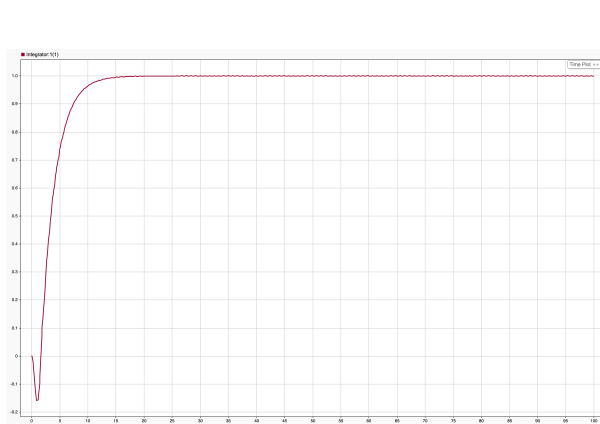
$$y_d(t) = \cos(0.2t) \tag{27}$$

$$z_d(t) = 1.5 \tag{28}$$

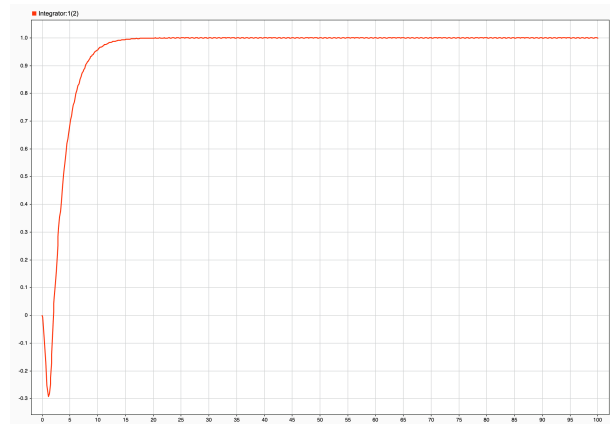
The quadrotor successfully tracked the circular trajectory.

7 Simulation Plots

7.1 Simulation 1: Set-Point Stabilization



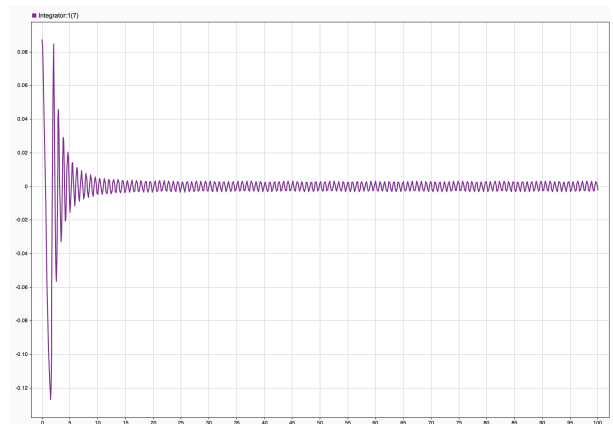
(a) X Position



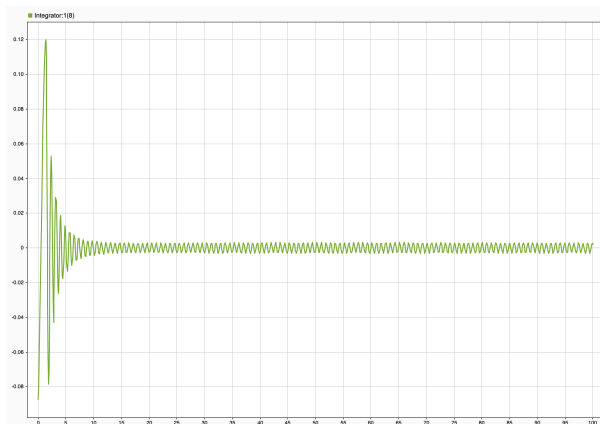
(b) Y Position



(c) Z Position



(d) Roll (ϕ)

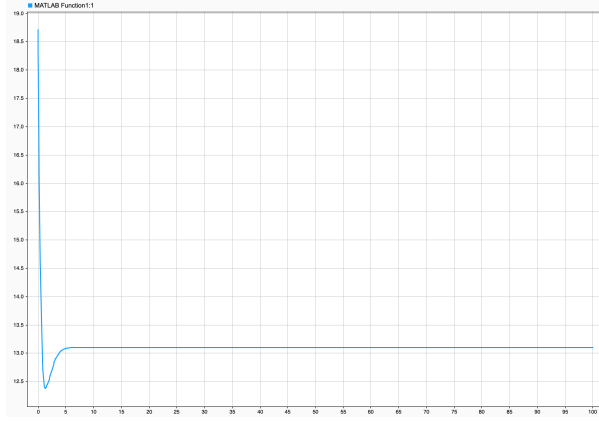


(e) Pitch (θ)

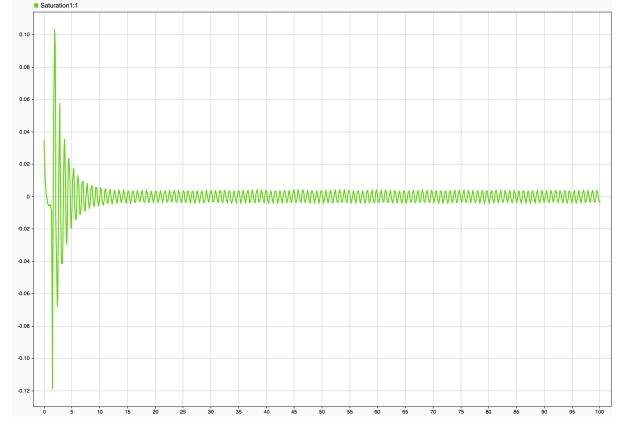


(f) Yaw (ψ)

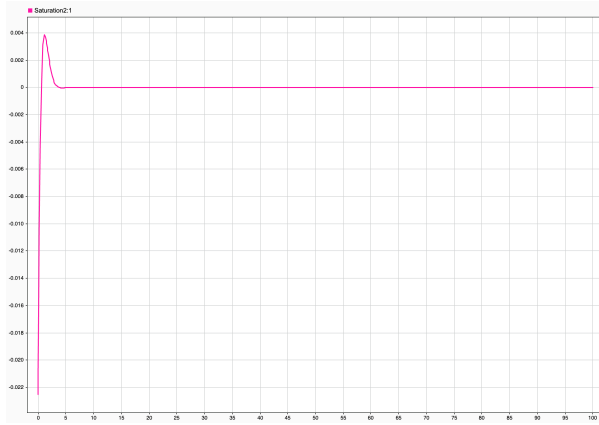
Figure 1: Simulation 1: Position and Attitude States



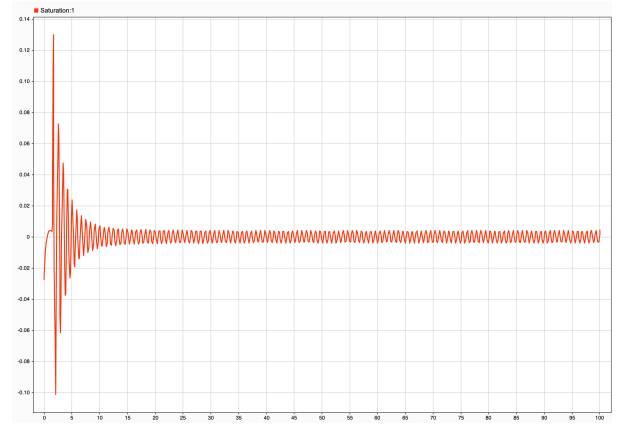
(a) U_1



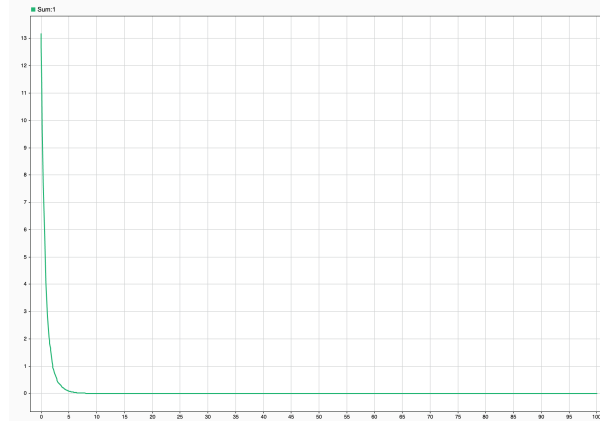
(b) U_2



(c) U_3



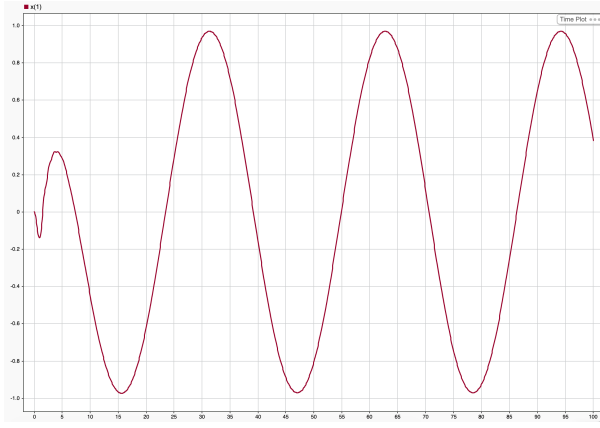
(d) U_4



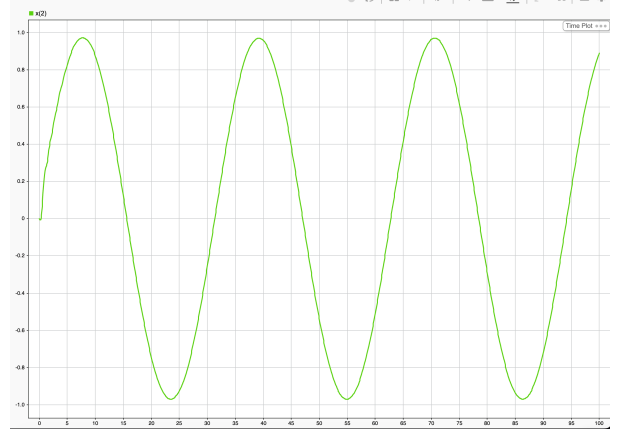
(e) Total Lyapunov V

Figure 2: Simulation 1: Control Inputs and Lyapunov Function

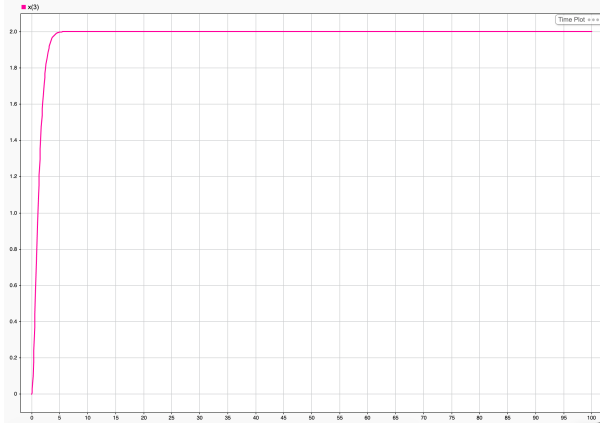
7.2 Simulation 2: Trajectory Tracking



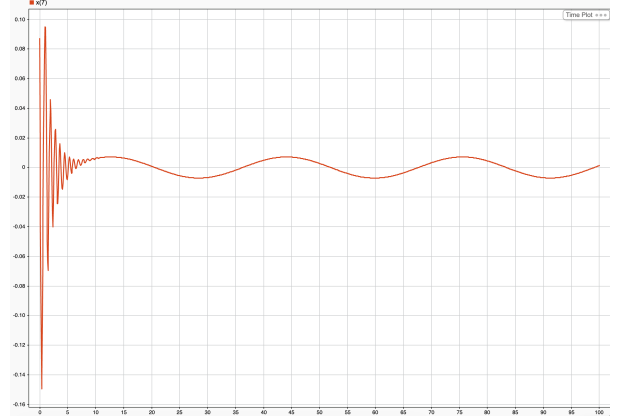
(a) X Position



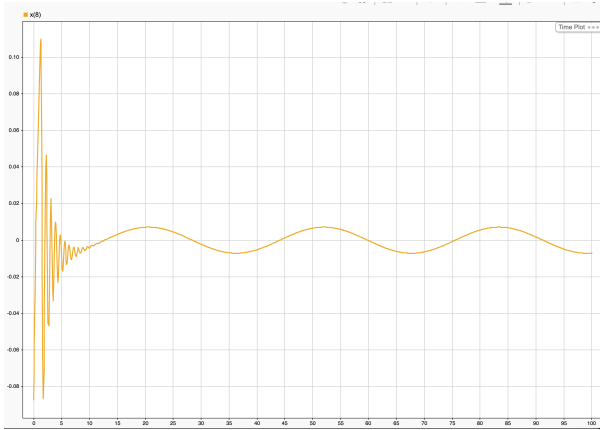
(b) Y Position



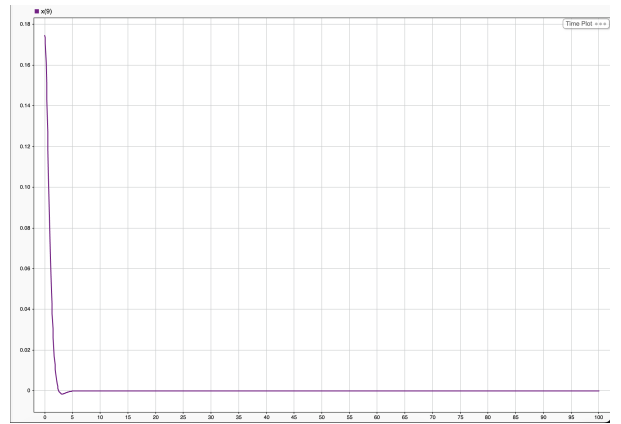
(c) Z Position



(d) Roll (ϕ)

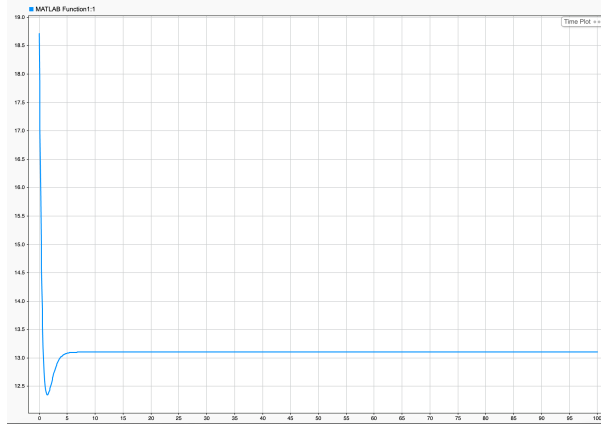


(e) Pitch (θ)

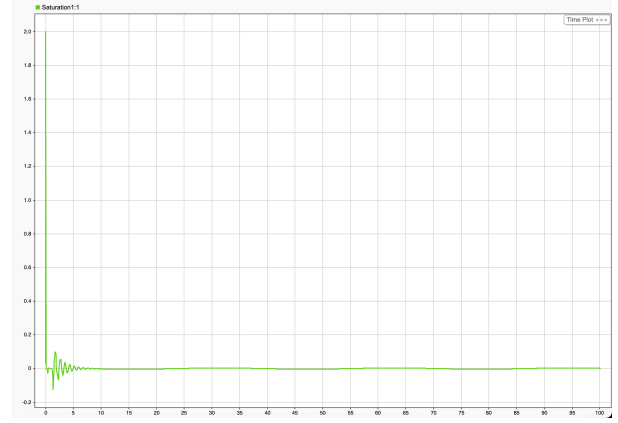


(f) Yaw (ψ)

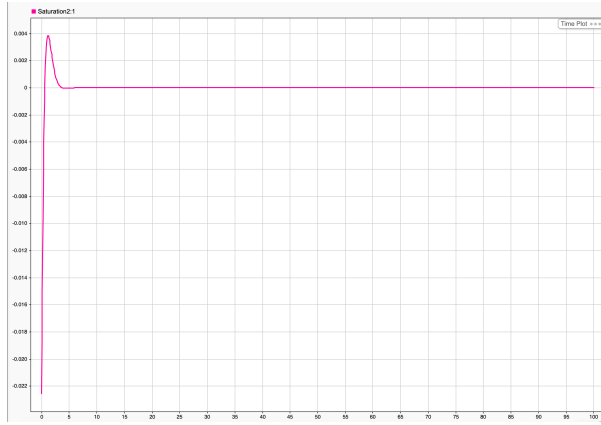
Figure 3: Simulation 2: Position and Attitude States



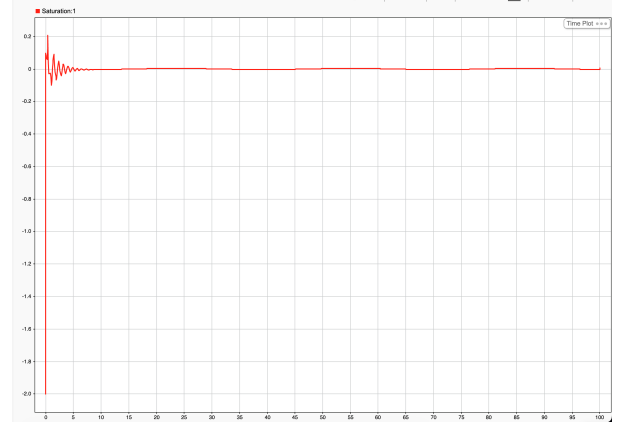
(a) U_1



(b) U_2



(c) U_3



(d) U_4



(e) Total Lyapunov V

Figure 4: Simulation 2: Control Inputs and Lyapunov Function

Implementation Details

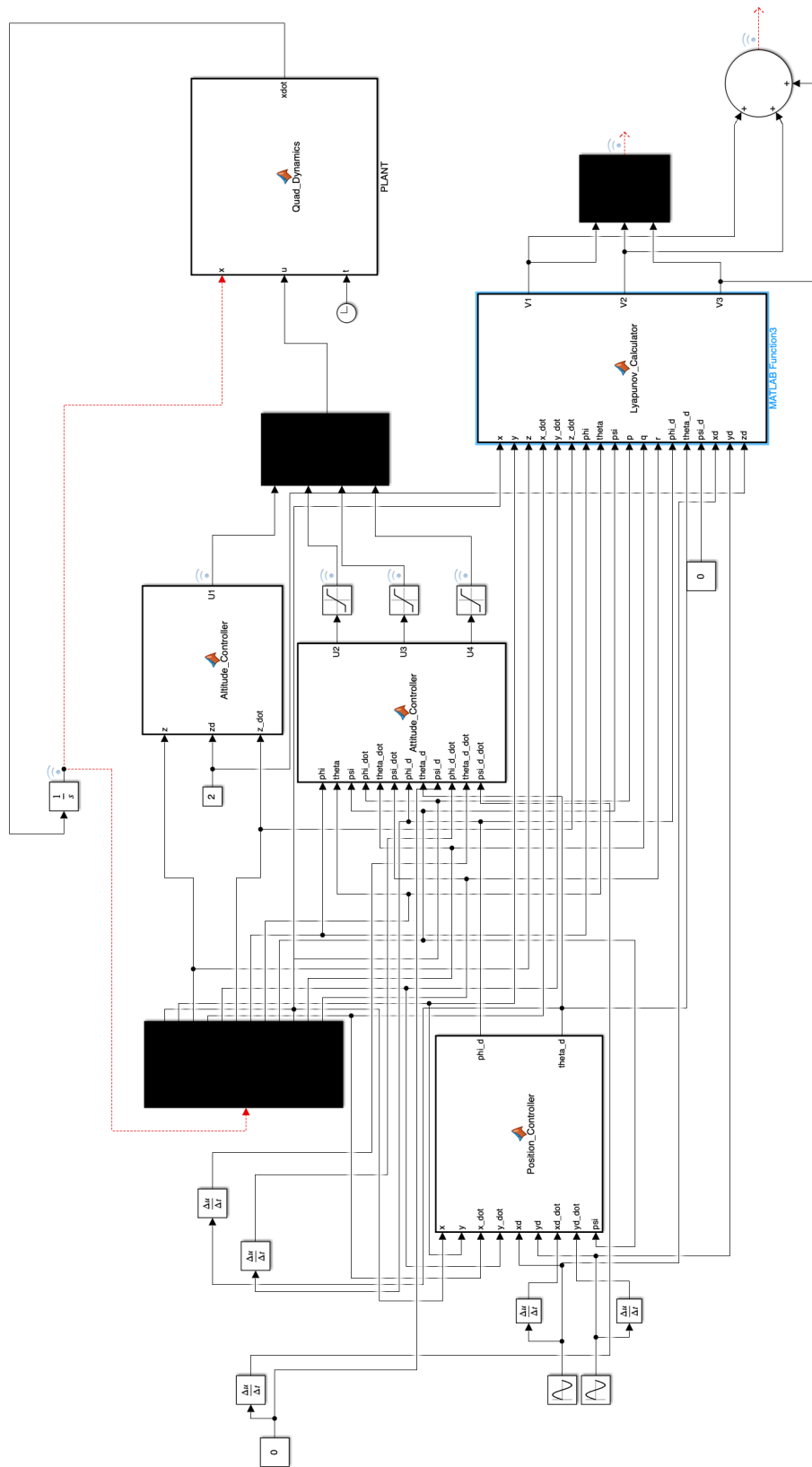


Figure: Complete Simulink Block Diagram of the Quadrotor Control System

8 Conclusion

The nonlinear backstepping controller ensures stability of the quadrotor through Lyapunov based design. Simulation results validate both set-point stabilization and trajectory tracking performance.